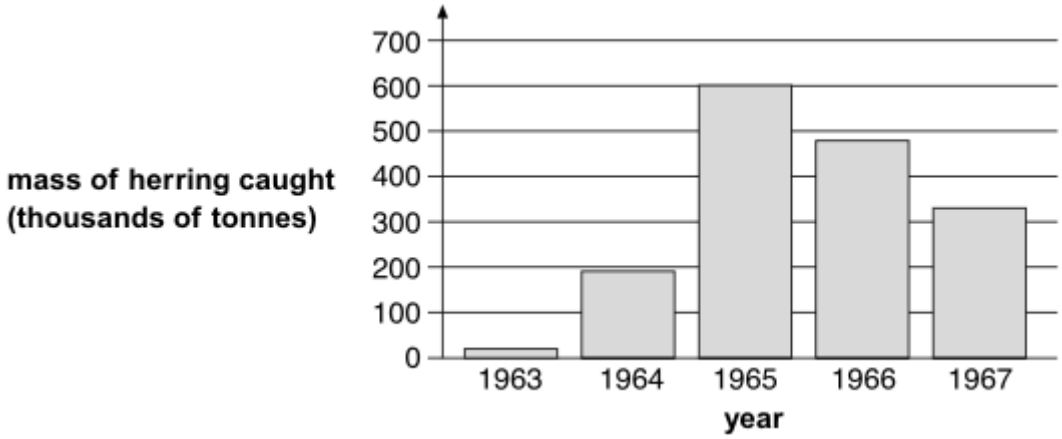
Q1. The table below shows the number of boats used for catching herring fish in the Norwegian Sea between 1963 and 1967.

year	number of fishing boats
1963	16
1965	284
1967	326

The bar chart below shows the total mass of herring caught in the Norwegian Sea between 1963 and 1967.



Use the information above to help you answer parts (a) (i), (ii) and (iii).

(a) (i) Why did the mass of herring caught increase between 1963 and 1965?

.....

.....

1 mark

(ii) Suggest why the mass of herring caught decreased between 1965 and 1967.

.....

.....

1 mark

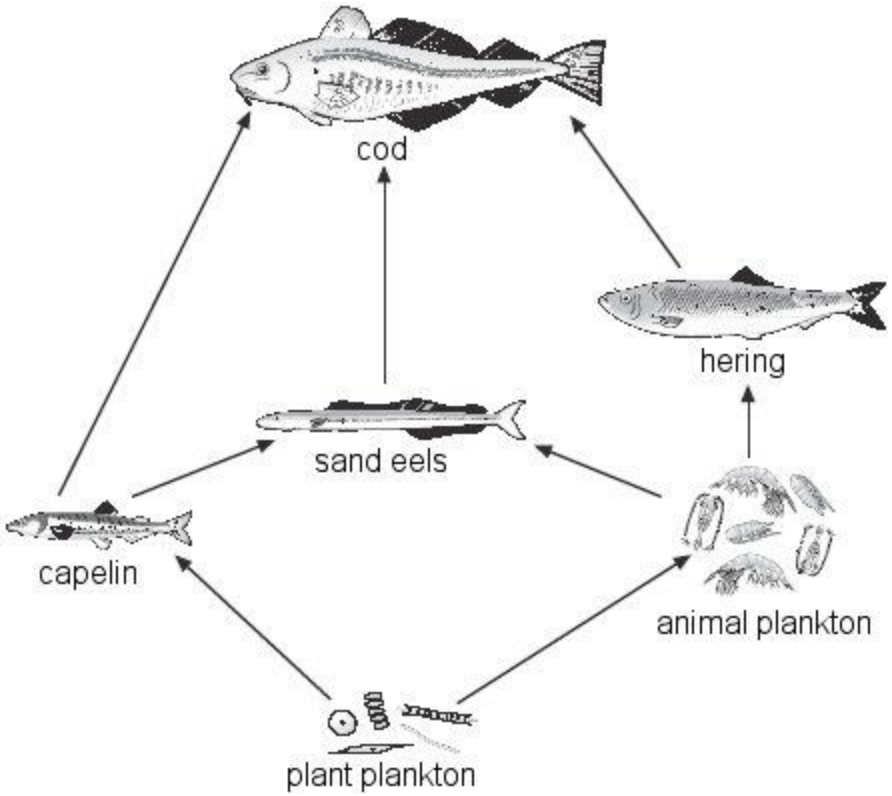
(iii) Herring **cannot** breed until they are four years old.
Fishing for herring was banned in the Norwegian Sea from 1972 to 1976.
Suggest **one** reason why fishing for herring was banned **for this period**.

.....

.....

1 mark

(b) The diagram below shows a food web in the Norwegian Sea.



not to scale

(i) How could a decrease in the number of herring cause a **decrease** in the number of sand eels?

.....

.....

1 mark

(ii) How could a decrease in the number of herring cause an **increase** in the number of sand eels?

.....

.....

1 mark

maximum 5 marks

Mark schemes

Q1.

(a) (i)

- more fishing boats
  accept 'more people were fishing'
  accept 'more fishing'
  accept 'more boats'
  'more being caught' is insufficient

1 (L5)

(ii) any **one** from

- overfishing
  accept 'too many fish were caught'
  'lots of fish were caught' is insufficient
  accept 'few fish or herring were left in the sea'
  *do **not** accept 'no herring **or** fish left'*
  *do **not** accept 'too many boats'*
- fish were caught before they could breed
- fish were not allowed time to breed
  accept 'fish were smaller'
  'the herring are too young' is insufficient
  'less being caught' is insufficient

1 (L5)

(iii) any **one** from

- to allow numbers to recover **or** increase
  *do **not** accept 'they were becoming extinct'*
- to allow more herring to breed
  accept 'so the herring would be old
  'to allow herring to be born' is insufficient
- to allow more herring to mature
  *enough **or** mature enough to breed'*
  accept 'it is the breeding time'
  'there were fewer herring left' is insufficient

1 (L5)

(b) (i) any **one** from

- cod eat sand eels instead of herring
  'cod have to eat something else' is insufficient
- cod eat more sand eels
  'cod eat sand eels' is insufficient
- cod eat capelin instead of herring so there is less food for sand eels
  accept 'cod would eat more capelin'
  accept 'cod would eat the sand eels' food'
  'less food for sand eels' is insufficient as it implies that sand eels eat herring

'because the cod only had two choices' is insufficient

1 (L6)

(ii) any **one** from

- there would be more animal plankton

  accept 'more plankton'

  *'they will have more to eat **or** more food' is insufficient*

- fewer cod

  'the animal plankton would get bigger' is insufficient
  'less food for cod' is insufficient

1 (L6)

[5]